

2.1 (1) $\pi_{\text{person_name}}$

(2) $\pi_{\text{person_name, company_name}}$

(3) $\pi_{\text{company_name}}$

2.6 (1) $\pi_{\text{person_name}} (\sigma_{\text{city} = \text{"Miami"}} (\text{employee}))$

(2) $\pi_{\text{person_name}} (\sigma_{\text{salary} > 10000} (\text{works}))$

(3) $\pi_{\text{person_name}} (\sigma_{\text{city} = \text{"Miami"}} (\text{employee}) \bowtie \sigma_{\text{salary} > 10000} (\text{works}))$

2.7 (1) $\pi_{\text{branch_name}} (\sigma_{\text{branch_city} = \text{"Chicago"}} (\text{branch}))$

(2) $\pi_{\text{ID}} (\text{borrower} \bowtie \sigma_{\text{branch_name} = \text{"Downtown"}} (\text{loan}))$

2.15 (1) $\pi_{\text{loan_number}} (\sigma_{\text{amount} > 10000} (\text{loan}))$

(2) $\pi_{\text{ID}} (\text{depositer} \bowtie \sigma_{\text{balance} > 6000} (\text{account}))$

(3) $\pi_{\text{ID}} (\text{depositer} \bowtie \sigma_{\text{balance} > 6000 \wedge \text{branch_name} = \text{"Uptown"}} (\text{account}))$